LESSON PLAN

AIRCRAFT ELECTRICAL EQUIPMENT, POWER GENERATION AND DISTRIBUTION (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	28
Total No. of Lesson Plans	35	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction – accelerometers and fatigue meters
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction – accelerometers and fatigue meters.
5. Specific Objectives	15.1.11.1 Trainee will be able to explain accelerometers and fatigue meters.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of instrument diagrams and operational principles; Safety: No live electrical connections; Supervision: Instructor monitors understanding of theoretical concepts.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an accelerometer in aircraft systems?
(b) How does a fatigue meter assist in maintenance scheduling?
(c) Where are accelerometers typically located on an aircraft?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and working principle of accelerometers • Types of accelerometers used in aviation • Applications of accelerometers in flight control • Basic construction and sensing elements • Signal processing and output mechanisms	L & Ds	PPT, IB	15 min
(b)	• Definition and working principle of fatigue meters • Cycle counting methods in fatigue analysis • Integration with aircraft maintenance programs • Data logging and trend analysis features • Limitations and error sources in fatigue meters	L & Ds	PPT, PDF	20 min
(c)	• Comparison between accelerometers and fatigue meters • Interdependence in structural health monitoring • Future trends in sensor technology • Regulatory requirements for instrument installation • Troubleshooting common instrument failures	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how an accelerometer detects vibration levels in an aircraft structure.
(b) Describe the role of a fatigue meter in predicting component lifespan.
(c) Identify two key differences between accelerometers and fatigue meters.

12. Home Assignment and Follow Up	Prepare a brief note on how the data from accelerometers and fatigue meters will be utilized in the next lesson regarding fatigue meter operation (LP 29).

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Fatigue meter operation.	Refer to the master syllabus reference list	LP 29

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min